

ASSIGNMENT ON CSE-260

Submitted to: Dewan Ziaul Karim (DZK) Sir

Submitted by: Sheikh Md. Tahmid Hossain

ID: 22299522

Section: 10

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Answer to the Question No.1

(a)

$$(A+B)(A+\bar{B})(\bar{A}+C)$$

$$= A(B+\bar{B})(\bar{A}+C)$$

$$= A(1)(\bar{A}+C)$$

$$= A(\bar{A}+C)$$

$$= A\bar{A} + AC$$

$$= AC$$

(b)

$$A'B'C + (A+B+C)' + A'B'C'D$$

$$= A'B'C + A'B'C' + A'B'C'D$$

$$= A'B'(C+C') + A'B'C'D$$

$$= A'B' + A'B'C'D$$

$$= A'B'(1+C'D)$$

$$= A'B'$$

$$= (AB)'$$

(c)

$$\begin{aligned} & XY + X(Y+Z) + Y(Y+Z) \\ &= XY + YX + XZ + Y(Y+Z) \\ &= XY + XZ + YY + YZ \\ &= XY + Y + XZ + YZ \\ &= Y(1+X) + XZ + YZ \\ &= Y + YZ + XZ \\ &= Y(1+Z) + XZ \\ &= Y + XZ \end{aligned}$$

(d)

$$\begin{aligned} & (AB' C + BD) + A'B' C \\ &= (AB' C + AB' BD + A'B' C) \\ &= (AB' C + A'B' C) \\ &= AB' CC + A'B' C \\ &= AB' C + A'B' C \\ &= B' C (A + A') \\ &= B' C \end{aligned}$$

Answer to the Question No. 2

Let,

$$F = (x' + y + z')(x' + y')(x' + z)$$

Duality

$$x'y'z', x'y', x'z$$

↓ ↓ ↓

$$x'y'z' + x'y' + x'z$$

After complementing each literal,

$$x'y'z + xy + xz'$$

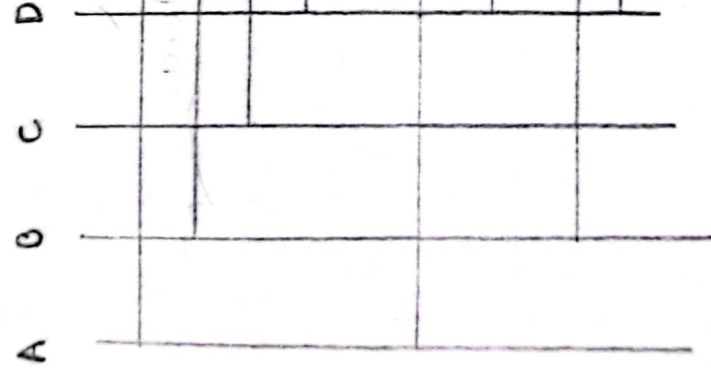
$$\therefore F' = x'y'z + xy + xz' \quad \text{Ans}$$

Answer to the Question No.3

Given,

$$F(A, B, C, D) = (A'B'C'D' + A'D + CB + D')$$

$$= (A'B'C'D' + A'D + (B + D'))$$



$(A'B'C'D')$

$(A'D)$

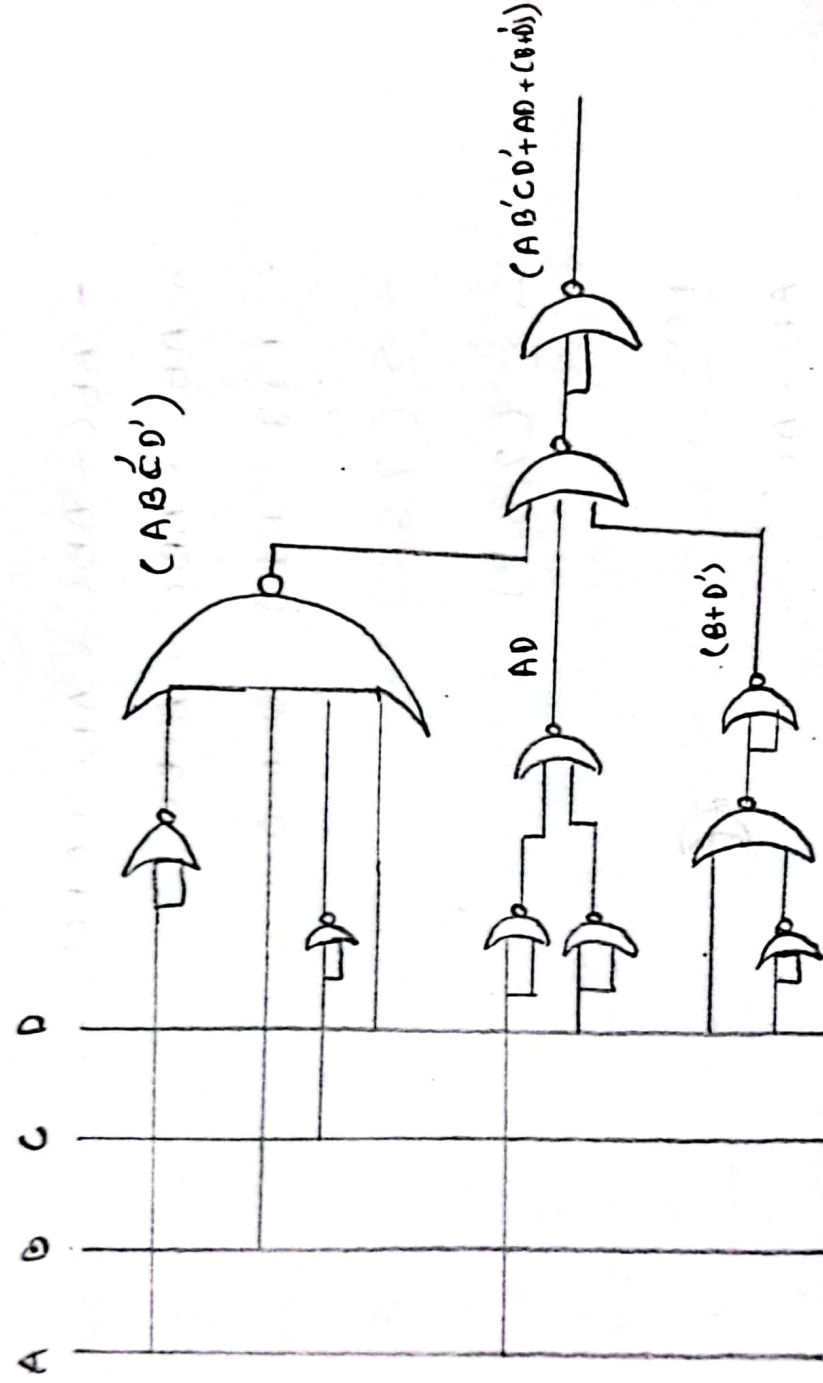
$(B + D')$

$(A'B'C'D' + A'D + (B + D'))$

Answer to the Question No. 4

Given.

$$F(A, B, C, D) = (AB'CD' + AD + (B+D'))$$



Answer to the Question No. 5

(a)

$$F(A, B, C) = AB + BC' \quad (\text{SOP})$$

$$(C + C') AB + BC' (A + A')$$

$$\rightarrow ABC + ABC' + ABC + A'BC$$

$$\rightarrow ABC + ABC' + A'BC$$

$$\rightarrow \begin{array}{ccc} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}$$

$$\rightarrow \Sigma (7, 6, 2)$$

$$\rightarrow \Sigma (2, 6, 7)$$

POS

$$AB + BC'$$

$$= BC(A + C')$$

$$= (B + AA')(A + B' + C')$$

$$= (A + B)(A' + B)(A + B + C')(A + B' + C')$$

$$= (A + B + CC')(A' + B + CC')(A + B + C')(A + B' + C')$$

$$= (A + B + C)(A + B + C')(A' + B + C)(A' + B' + C')(A + B + C)$$

$$= (A + B + C)(A + B + C')(A' + B + C)(A' + B' + C')$$

$$= 000, 001, 100, 101, 011$$

$$= \Pi (0, 1, 3, 4, 5)$$

(b)

$$F(A, B, C, D, E) = A + B'CO'$$

SOP

$$= A(B+B') + (A+A')BC'D'$$

$$= AB(C+C') + AB'(C+C') + AB'CD'(E+E') +$$

$$A'B'CD'(E+E')$$

$$= ABC(D+D') + ABC'(C+D') + ABC(D+D') + AB'CD'(D+D')$$

$$+ AB'CD'DE + A'B'CD'DE + A'B'CD'DE'$$

$$= ABCD(E+E') + ABCD'(E+E') + ABC'D(E+E') + ABC'D'(E+E')$$

$$+ ABC'D'(E+E') + AB'CD(E+E') + AB'CD'(E+E')$$

$$+ AB'CD'(E+E') + AB'CD'D'(E+E') + AB'CD'D'E +$$

$$AB'CD'D'E'$$

$$= ABCDE + ABCDE' + ABCD'E + ABCD'E' + ABC'DE$$

$$+ ABC'DE + ABC'D'E' + ABC'DE + ABC'DE' + AB'CD'E$$

$$+ AB'CD'E' + AB'CD'E + AB'CD'E' + AB'CD'E' +$$

$$AB'CD'D'E' + AB'CD'D'E + AB'CD'D'E'$$

$$= 11111 + 11110 + 11101 + 11100 + 11011 + 11010 + 11001$$

$$+ 11000 + 10111 + 10110 + 10101 + 10100 + 10011$$

$$+ 10010 + 10001 + 10000 + 00101 + 00100$$

$$= \sum 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21,$$

$$20, 19, 18, 17, 16, 15, 4)$$

$$= \sum (4, 5, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,$$

$$26, 27, 28, 29, 30, 31)$$

POS

$$A + B' C D'$$

$$= (A+B')(A+CD')$$

$$= (A+B'+CC')(A+C+BB')(A+D'+EE')$$

$$= (A+B'+C+DD')(A+B'+C'+DD')(A+B+C+DD')$$

$$(A+B'+C+DD')(A+D'+E+BB')(A+D'+E'+BB')$$

$$= (A+B'+C+D+EE')(A+B'+C+D'+EE')(A+B'+C'+D+EE')$$

$$(A+B'+C+D'+EE')(A+B'+C+D+EE')(A+B'+C+D'+EE')$$

$$(A+B'+D'+E+CC')(A+B'+D'+E+CC')$$

$$(A+B'+D'+E'+CC')(A+B'+D'+E'+CC')$$

$$= (A+B'+C+D+E)(A+B'+C+D+E')(A+B'+C+D'+E)$$

$$(A+B'+C+D'+E)(A+B'+C'+D+E)(A+B'+C'+D+E')$$

$$(A+B'+C'+D'+E)(A+B'+C'+D'+E')(A+B'+C'+D+E)$$

$$(A+B'+C+D+E')(A+B'+C+D'+E)(A+B'+C+D'+E')$$

$$(A+B'+C+D+E)(A+B'+C+D'E)(A+B'+C+D'+E)$$

$$\rightarrow w y v x + w y v x' + w y v' x + w y v' x'$$

$$w y v x (z+z') + w y v x' (z+z') + w y v' x (z+z') + w y v' x' (z+z') + w y v' x' (z+z')$$

$$\rightarrow w y v x z + w y v x z' + w y v' x z + w y v' x z' + w y v' x' z + w y v' x' z'$$

Now for wx

$$w x (v+v') + v w x + v' w x$$

$$v w x (y+y') + v' w x (y+y')$$

$$\rightarrow v w x y + v w x y' + v' w x y + v' w x y'$$

$$\rightarrow w x y z + v w x y z + v w x y z' + v w x y' z + v' w x y z + v' w x y' z + v' w x y' z' + v' w x y' z'$$

Now for x'y

$$x' y (v+v') + v x' y + v' x' y$$

$$v x' y (w+w') + v (x' y (w+w'))$$

$$\rightarrow v w x' y + v w x' y + v' w x' y + v' w x' y$$

$$\rightarrow v w x' y z + v w x' y z + v w x' y' z + v w x' y' z + v' w x' y z + v' w x' y' z + v' w x' y' z + v' w x' y' z$$

$$z' y' w' v$$

= (c)

$$F(v, w, x, y, z) = wy + wx + x'y$$

SOP

$$wy + wx + x'y$$

So, using v as the first term and onwards with w, x, y, z

$$wy(v+v') + wx(v+v') + x'y(v+v')$$

$$= vwy(x+x') + v'wy(x+x') + vwx(y+y') + v'wx(y+y') \\ + (x'x'y(w+w') + v'x'y(w+w'))$$

$$= vwxxy(z+z') + vwx'y(z+z') + (v'wx'y(z+z') \\ + v'wx'xy(z+z')) + vwxxy'(z+z') + vwx'y'(z+z') \\ + (v'wxxy(z+z') + v'wx'y'(z+z')) + vwx'xy'(z+z') \\ + vx'x'wy(z+z') + vx'x'wy'(z+z') + v'x'x'w'(z+z')$$

$$= vwxxyz + vwx'yz' + vwx'yz + vwx'yz' + v'wxxyz + \\ v'wx'yz' + v'wx'yz + v'wx'yz' + vwx'y'z + vwx'y'z' \\ + v'wx'yz + v'wx'yz' + vwx'x'yz + vwx'x'yz' \\ + v'wx'x'yz + v'wx'x'yz'$$

$$= 11111 + 11110 + 11011 + 11010 + 11010 + 01111 + 01110$$

$$01011 + 01010 + 11101 + 11100 + 01101 +$$

$$01100 + 10011 + 10010 + 00011 + 00010$$

$$= 31 + 30 + 27 + 26 + 15 + 14 + 11 + 10 + 29 + 28 + 13 + 12 + 19 + 18 + 3 + 2$$

$$= \sum (2, 3, 10, 11, 12, 13, 14, 15, 18, 19, 26, 27, 28, 29, 30, 31)$$

POS

$$wx + wx' + x'y$$

[We need to apply

distributive law

$$= w(y+x) + x'y$$

where necessary

$$= (w + x'y)(y+x + x'y)$$

$$xc + yz = (yc + y)$$

$$(y-x+2)$$

$$= (w + x')(w+y)(y+x+x')(y+x+y)$$

$$= (w+x')(w+y)(x+y)$$

$$= (w+x' + vv')(w+y + w')(x+y+w')$$

$$= (w+v+x')(v'+w+x')(v+w+y)(v'+w+y)$$

$$(v+x+y)(v'+x+y)$$

$$= (v+w+x' + yy')(v+w+x' + y' + zz')(v'+w+x' + y + zz')$$

$$(v'+w+x' + zz')(v+w+x+y+zz')(v+w+x'+y+zz')$$

$$\begin{aligned}
&= (v+w'+x+y+zz') (v'+w+x+y+zz') (v'+w+x+y+zz') \\
&= (v+w+x'+y+z) (v+w+x'+y+z') (v+w+x'+y'+z) \\
&\quad (v+w+x'+y'+z') (v'+w+x'+y+z) (v'+w+x'+y'+z') \\
&\quad (v'+w+x'+y'+z) (v'+w+x'+y'+z') (v+w+x+y+z) \\
&\quad (v+w+x+y+z') (v'+w+x+y+z) (v'+w+x+y+z') \\
&\quad (v+w'+x+y+z) (v+w'+x+y+z') (v'+w'+x+y+z) \\
&\quad (v'+w'+x+y+z')
\end{aligned}$$

$$\begin{aligned}
&= 00100, 00101, 00110, 06111, 10100, 10101 \\
&\quad , 10110, 10111, 00000, 00001, 10000, 10001 \\
&\quad , 01000, 01001, 11000, 11001
\end{aligned}$$

$$= \Pi (4, 5, 6, 7, 20, 21, 22, 23, 0, 1, 16, 17, 8, 9, 24, 25)$$

$$= \Pi (0, 1, 4, 5, 6, 7, 8, 9, 16, 17, 20, 21, 22, 23, 24, 25)$$

$$\begin{aligned}
 & (A+B+c+d'+e') (A+B+c+d'+e) (A+B+c'+d'+e) \\
 & (A+B'+c+d'+e) (A+B'+c'+d'+e) (A+B+c+d'+e') \\
 & (A+B+c'+d'+e') (A+B'+c+d'+e') (A+B+c'+d'+e')
 \end{aligned}$$

$$\begin{aligned}
 & = (A+B'+c+d'+e) (A+B'+c+d'+e) (A+B'+c+d'+e) \\
 & (A+B'+c+d'+e') (A+B'+c'+d'+e) (A+B'+c'+d'+e') \\
 & (A+B'+c'+d'+e) (A+B'+c'+d'+e') (A+B+c+d'+e) \\
 & (A+B+c+d'+e') (A+B+c+d'+e) (A+B+c+d'+e') \\
 & (A+B+c'+d'+e) (A+B+c'+d'+e)
 \end{aligned}$$

$$\begin{aligned}
 & = 0100, 01001, 01010, 01011, 01100, 01101, \\
 & 01110, 01111, 0000000, 00001, 00010, 00011, \\
 & 00110, 00111
 \end{aligned}$$

$$= \pi (0,1,2,3,6,7,8,9,10,11,12,13,14,15)$$